

Cedula e-Service Portal

Securing Capstone

Presented to the Faculty of the
College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled Cedula e-Service Portal prepared and Submitted by Karen Allones, Alh Jane Baricuatro, Ma Lyn Gayuna, who are hereby recommended for corresponding final evaluation.

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To all those who, in one way or another, supported us, We extend our sincere appreciation and gratitude

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ABSTRACT

This project addresses the inefficiencies of the current Community Tax Certificate (CTC) Cedula application process, it has long processing times and complex procedures. The proposed Cedula e-Service portal is a user-friendly and web-based solution designed to optimize CTC procurement. Utilizing OTP verification, the system enhances security and efficiency. Applicants provide personal information, which is verified via OTP authentication, adding multiple layers of security against fraud. After completing an online form and scheduling an appointment, payment and thumb mark is processed onsite. A confirmation email with a verification link for the appointment further secures the transaction. This system significantly reduces processing time to under 10 minutes or less, minimizing unnecessary trips to municipal offices while maintaining the security. The integration of secure verification protocols ensures a more efficient and trustworthy CTC application process.

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CHAPTER I

Introduction

In today's digital age, a demand and need for easy, accessible government service is important. Traditionally, acquiring a cedula often involves a time consuming process of manual paperwork and long queues, which often causes citizens to become frustrated and inefficient.

A cedula is a basic requirement for many important tasks like applying for a job, starting a business, and processing important paperwork. But the old way of processing a cedula is inconvenient. This project will be a stepping stone to show how technology can make life easier for our citizens and to initiate e-governance services.

This study focuses on designing, developing and implementing a cedula e-service portal. We aim to modernize the cedula application and issuance process by providing a secure and user-friendly platform for the citizens.

Background of the Study

The cedula is a fundamental document for the citizens in our country to serve as a vital role in accessing essential services like applying for a job, starting a business and processing important paperwork. However, the process of obtaining a cedula is often burdensome and time-consuming.

I have a personal experience with this process, I have to wait in line and manually fill out a form as it is the basis to input my information in a cedula. This encounter highlights the need for improvement, efficient and user-friendly platform for cedula processing.

The inefficiency of the current procedure is a burden for some individuals. It affects the government services and satisfaction of the citizens. It is a challenge for someone who has limited mobility or residing in remote areas. The paper-based processes have a possibility for fraud, data privacy and errors as it may be a big concern for the integrity of a Cedula system.

This encounter made me want to learn more about digital transformation on the newest cedula platform. We investigate the means in which e-service portals contribute to the success of information management, motivated by the results and initiatives of other nations' e-governments. With the goal of providing

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government services more effectively, an e-service portal has the ability to improve security, increase accessibility, and streamline procedures.

This study aims to test the design, improvement and implementation of the cedula e-service portal and what benefits and challenges it may face. To analyze its impact on efficiency, accessibility and security, this study aims to give a contribution to the municipality of Leon and give away for a more efficient and modern approach to cedula processing and identity management.

Statement of the Problem

The efficiency, accessibility and security of government existing cedula issuance processes are all affected by serious issues. Among these difficulties are:

- **Time-consuming Processing:** the paper-based process often needs a multiple visit in the government office resulting in insignificant inconvenience for citizens.
- **Limited Accessibility:** the physical processing in the government office is a barrier for some individuals who are siding in remote areas, which may worsen the challenges in processing a cedula.

- **Possibility for fraud and error:** paper-based processes are prone to errors, loss of document and potential for fraud that will compromise the integrity and security of the cedula acquisition.

This study seeks to address these problems by developing a cedula e-service portal that integrates modernization on processing and opens a way for a more efficient and secure approach to identity management.

Objectives of the study

The main objective of the study is to design and develop a cedula e-service portal that improves the efficiency, integrity and security of the cedula processing. It aims to develop and evaluate a system to analyze the effectiveness of the e-service portal and to identify best practices for a successful implementation and to address the potential challenges it may face.

Significance of the study

The research holds a significant implication for individuals, contributing to a more efficient accessible and secure approach to cedula processing:

1. **Citizens:** it offers citizen convenience accessibility and increases security. It is a user-friendly platform, eliminating the need to visit and improve the service for those with disabilities or in remote areas.
2. **Government Agencies:** Reducing administrative burden, improving operational efficiency, enhancing their data management and improving the citizen engagement.
3. **Researcher/Proponent:** It will give them helpful information for changing how we do identity management and to give them insight about online government services and improve discussion on online government in giving better public services.

Scope and Limitations

This study focuses on the development of a cedula e-service portal to improve efficiency, integrity and security in processing cedula. The key features of the system include the target audience, the functionality of the system, accessibility and it will also address the limitations like data privacy and security, the maintenance and update and the resources and

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funding needed. This study will cover all the key aspects of this project:

1. **Target audience:** the target audience includes the citizens, the government and individuals who reside in remote areas.
2. **Functionality:** it encompasses the entire cedula issuance process, online application, submission, verification for payment processing on site.
3. **Accessibility:** To cater individuals with disability and those who reside in remote areas.
4. **Data privacy and security:** maintaining privacy and security, required monitoring, updates and compliance.
5. **Maintenance and updates:** maintenance and update are essential to ensure the system's reliability, security and integrity.

By implementing this key aspect cedula e service portal aims to provide a user friendly and security detail experience for citizens.

Limitations of the study

While the study aims to address the several aspects of cedula e-Service portal there are some limitations:

1. **Scope:** The system will focus specifically on the implementations of the cedula e-Service Portal to the municipality of Leon. This system is not applicable to other municipalities or regions that have a different demographics, infrastructure, or government system.
2. **Digital Literacy and Access:** It acknowledges the potential limitations to those who have access among citizens. This may cause biases, as those who have less digital knowledge or internet access are at a disadvantage.
3. **System Integration:** As it is a government, the integration of this system to the existing government system is difficult as it needs a lot of processing and approval. Technical difficulties or compatibility issues during the integration could affect the portal performance and it may compromise the data.
4. **On-site Processing:** The requirement for in person attendance for thumbprint can be one of the bottlenecks that this project will face. These findings may be limited by efficiency of the onsite processing.
5. **Security threats:** As it is known security is very important. The design of the system needs to be tested in security and vulnerability, and we lack security knowledge to test the security in a better way.

Definition of Terms

1. **Cedula (Community Tax Certificate):** A fundamental document required for various transactions in the Philippines. Serving as a proof of residency and payment of community tax.
2. **e-Service Portal:** An online platform that has access to government office, in this case, the application and processing of cedula
3. **OTP (One-Time Password):** A strong generated password used for secure user authentication adding a layer of security to the system
4. **Digital Literacy:** The ability of individuals to use and have knowledge of digital technologies, including computers, internet and mobile devices.
5. **System Integration:** The process of connecting the system to existing government databases and IT infrastructure to ensure seamless data and functionality.
6. **Agile Methodology:** An iterative and flexible approach to software development that emphasizes the planning design development testing deployment review and launch.
7. **Data Flow Diagram (DFD):** A graphical representation of the flow of data through a system, showing how data is processed and transformed.
8. **Entity Relationship Diagram (ERD):** A visual representation of the relationship between different data entities within a system, such as user, application and records.

Chapter II

Literature Review

The chapter explores relevant research and literature concerning the development of a Cedula e-Service Portal, focusing on secure online service implementation and e-government solution adoption. The research highlights the growing significance of secure and user-friendly e-government platforms, emphasizing the use of technology acceptance models, security considerations, and user-centric design in developing the Cedula e-Service Portal for efficient and secure community tax certificate acquisition. Cadeliña's 2021 study on real-time transaction processing systems (RTTPS) in e-government (Cadeliña, 2021) highlights the increasing need for online service delivery, particularly during pandemics, to minimize in-person interactions and efficiently deliver services like online tax payments. This emphasizes the importance of e-government for streamlined processes, enhanced data transparency, and improved service delivery. Cadeliña's research utilizes the Technology Acceptance Model (TAM) by Davis (1989), alongside the Theory of Reasoned Action (TRA), Theory of Planned Behavior (TPB), and Unified Theory of Acceptance and Use of Technology (UTAUT), to assess RTTPS acceptability and usability, providing a comprehensive framework for understanding user acceptance of e-government services. While focusing on community tax certificates, the study indirectly addresses security,

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emphasizing secure database management and data protection practices, including data encryption, strong authentication mechanisms, database security policies, and regular security audits and penetration testing. Cadeliña's findings demonstrate high user acceptance of the RTTPS, highlighting the potential for online solutions to improve service delivery and satisfaction when designed with user-centricity. The research also details the RTTPS development using the Rapid Application Development (RAD) Model, emphasizing iterative development, user feedback, and rapid functional system delivery. Furthermore, a comparison of the Enhanced Porter's Stemming Algorithm (EPSA) and a real-time transaction algorithm for retrieving taxpayer information highlights the importance of selecting appropriate search and retrieval algorithms for efficient data access.

Additionally, a study by Gant, Burley Gant, and Gant (2002) investigated the functionality of web portals across all 50 US states (Gant, Burley Gant, & Gant, 2002). This study found that most were in their early stages of development, offering limited access to government services. The study emphasized the importance of system integration to provide advanced transaction capabilities, a feature largely absent from most portals. The researchers identified strategic IT approaches and e-government friendly laws as crucial factors for high-functioning portals. This research highlights the need for the Cedula e-Service portal to prioritize system integration, adopt a cohesive IT strategy, and ensure legal compliance to create a secure and

effective service. The study further defined four key dimensions of web portal functionality: openness, customization, usability, and transparency. This framework serves as a valuable tool for evaluating the Cedula e-Service portal design and implementation, ensuring it meets user expectations and fosters trust and engagement. Additional insights from the study, including the trend of organizing e-services around events, the importance of a web-savvy population, and the integration of emerging technologies, provide further guidance for developing a user-friendly and effective online service.

References

- Cadeliña, F. A. (2021).** Efficacy of Real-Time Transaction Processing System. *Journal of Communication and Computer*, 18(1), 19-25.
- Gant, J. P., Burley Gant, D., & Gant, J. (2002).** Web portal functionality and state government e-service. In *Proceedings of the 35th Hawaii International Conference on System Sciences*.

Chapter III

Methodology

This chapter outlines the research methodology that will be employed in the development of the Cedula e-Service Portal. This research aims to design and implement a user-friendly, secure, and efficient online platform for processing and managing Cedula applications, leveraging modern technologies like cloud computing and data analytics. The methodology will detail the research approach, data collection methods, system design, tools, and techniques used to achieve this objective.

Data Collection Techniques

Method	Target Participants	Data Type	Purpose
Surveys	Government Officials, Citizens with Cedula Experience	Quantitative	To gather data on current experiences, challenges, and expectations for the Cedula e-Service Portal.

Interviews	Small groups of respondents including government officials, and citizens	Qualitative	To gain deeper insights into the challenges, workflows, and potential benefits of an e-Service Portal.
Focus Groups	Selected individuals with expertise in government services, or Cedula processing	Qualitative	To gather diverse perspectives on the system's functionality, usability, and potential challenges.
Observations	Government offices processing Cedula applications	Qualitative	To understand the current process, identify inefficiencies, and pinpoint areas where an e-Service Portal could improve.
Document Analysis	Existing documents, reports, and case studies on online government services	Qualitative	To identify best practices, trends, and technological advancements for the Cedula e-Service Portal.

Research and Development Stage

The Research and Development (R&D) stage is crucial for the development of the Cedula e-Service Portal. During this stage, the development process will be guided by insights and feedback collected from respondents, such as government officials, IT professionals, and citizens who have experience with Cedula applications.

- **Data Analysis:** The R&D stage will begin with an analysis of data gathered from surveys and interviews. These responses will identify key challenges faced by individuals and agencies involved in the Cedula process.
- **System Design:** Based on the analysis, the system architecture will be designed. The architecture will incorporate modern technologies like cloud computing for data storage, security features for data protection, and user-friendly interfaces for seamless interactions.
- **Development and Testing:** The Cedula e-Service Portal will be built in stages, with regular testing and evaluation to meet performance standards. Feedback from respondents will be incorporated to refine and improve the system. Once a prototype is completed, user testing will be conducted to

ensure intuitiveness, user-friendliness, and efficiency in processing Cedula applications.

Software Development Life Cycle (SDLC)

The software development lifecycle (SDLC) is a structured approach used in software development of the "Cedula e-Service Portal" to develop a project that is completed successfully, effectively and efficiently. The SDLC will guide the development process from the initial planning to final launching.

Key features of Agile Methodology

- **Iteration:** The system are broken down into small, manageable increments, often called sprint, that are developed and tested in cycles to deliver working software frequently.
- **Collaboration:** Close collaboration between developers, users/clients, and advisers to ensure alignment with user needs.
- **Feedback:** Receive feedback from the stakeholders for improvement and ensure alignment with the user/client.
- **Flexibility:** Allowed changes in requirement even during the later stages of development.

The Software Development Life Cycle (SDLC) will guide the development process, ensuring the Cedula e-Service Portal meets user needs and aligns with insights provided by respondents. The SDLC will consist of the following stages:

1. **Planning and Requirement Analysis:** The initial phase involves gathering detailed requirements for the system. This will involve identifying the specific challenges and needs of the Cedula process, including user needs, security protocols, and data management. Through surveys, interviews, and discussions with government officials, IT professionals, and citizens, the research team will gather data on key features, desired functionalities, and technical requirements of the e-Service Portal.
2. **System Design:** The system's architecture and overall structure will be defined in the design phase. Feedback from respondents will guide the design of both the front-end (user interface) and back-end (server and database structure). The design will incorporate modern technologies like cloud computing and security measures, ensuring scalability and user-friendliness.
3. **Development:** The actual coding and implementation of the system will occur during the development phase. Developers will create the software using appropriate programming languages and tools, based on the designs. Continuous feedback from respondents will be incorporated throughout this stage to ensure the system aligns with user needs and expectations.

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4. **Testing:** Once the system is developed, thorough testing will be conducted, including functional testing to check all features are working correctly and performance testing to ensure the system can handle large amounts of data and usage. Respondents will participate in user acceptance testing (UAT), where they will test the system in real-world conditions to identify any issues or areas for improvement.
5. **Deployment:** After successful testing, the system will be deployed in a live environment. This phase involves installing and configuring the system for actual use, ensuring it is fully operational and accessible to users. Initial training will be provided to users to effectively operate the system.
6. **Maintenance and Support:** Regular updates, bug fixes, and additional features will be integrated based on user feedback and evolving needs.

Justification for selecting Agile Methodology

The decision to adopt Agile Methodology over other SDLC models such as waterfall, was based on several critical factors.

Criteria	Agile	Waterfall
Flexibility	High- embrace changes and allow for modification to requirements throughout the development	Rigid, sequential process making it difficult and costly to accommodate changes once completed
User Involvement	Continuous user involvement and collaboration	Limited user involvement that can lead to unmet expectations
Development Speed	Faster delivery through iterative sprint	Slower due to sequential phase
Risk Management	Early detection and resolution of issue	Risk may surface late in the process
Project suitability	Ideal for evolving requirements	Vest for well defined, static projects

System Architecture and Design

Overview

The cedula e-service portal was designed to improve cedula acquisition. The architecture follows three tier architecture to ensure a user-friendly, secure and efficient online platform. The architecture comprises three logical layers - presentation layer, application layer and data layer that work to deliver these services.

System Architecture

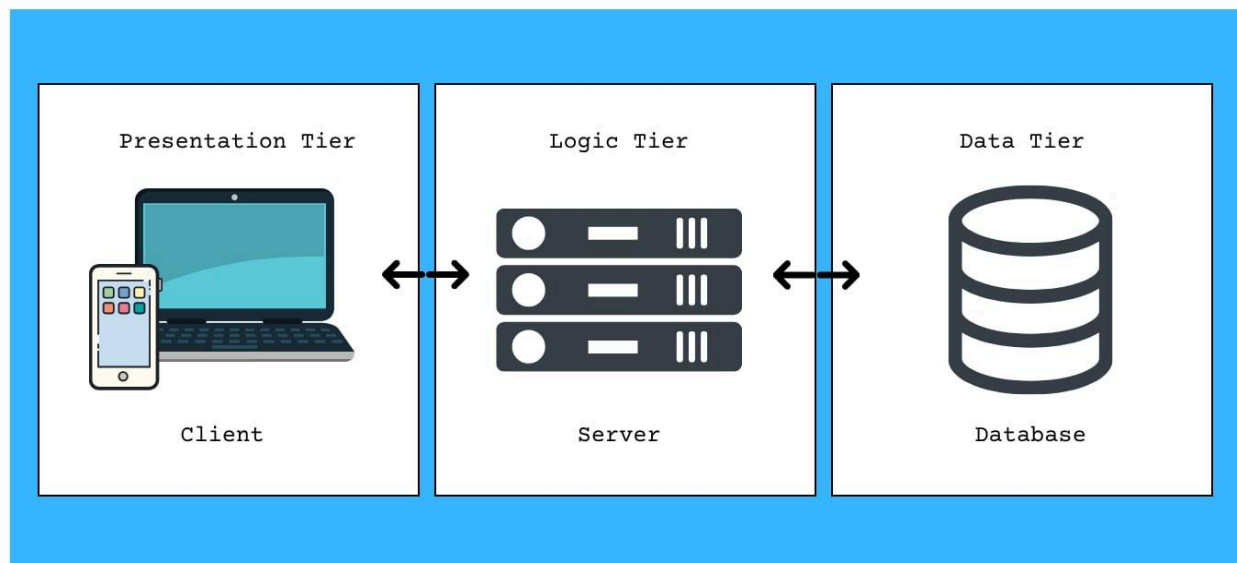


Figure 3.1 Three-tier architecture

The system architecture is divided into three main layer:

- 1. Presentation Layer (Presentation Tier):** This layer is where the citizens interact with the system. It includes web pages, mobile applications that allow users to fill out application form, online payment, tracking of the status of application.
- 2. Application Layer (Logic Tier):** This layer contains the logic of the system. It handles user authentication, form validation, data processing, payment processing, security and communication with other system components.
- 3. Data Layer (Data Tier):** This layer manages the storage and retrieval of data. It includes MySql Database for storing user data or information, application details, records and other data.

System Components and Modules

Landing Page

Figure 3.2 Serves as the first interaction of the user or the applicant

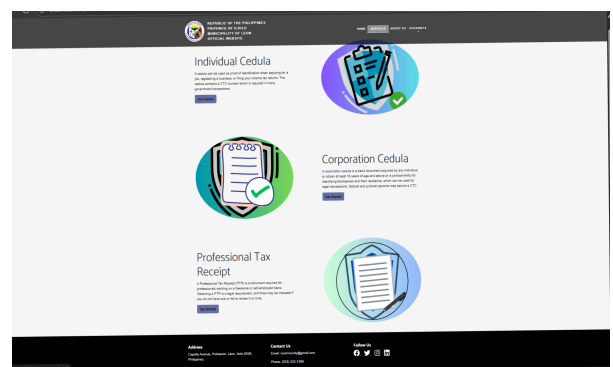
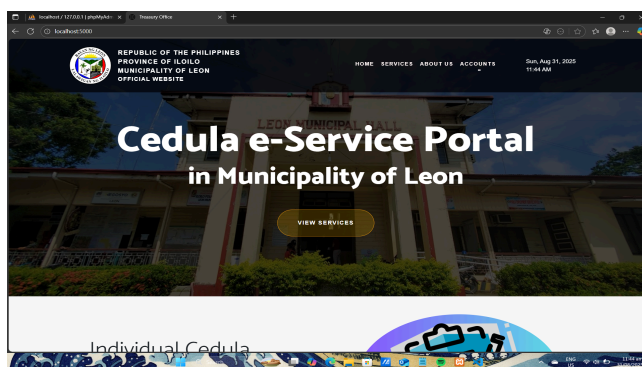


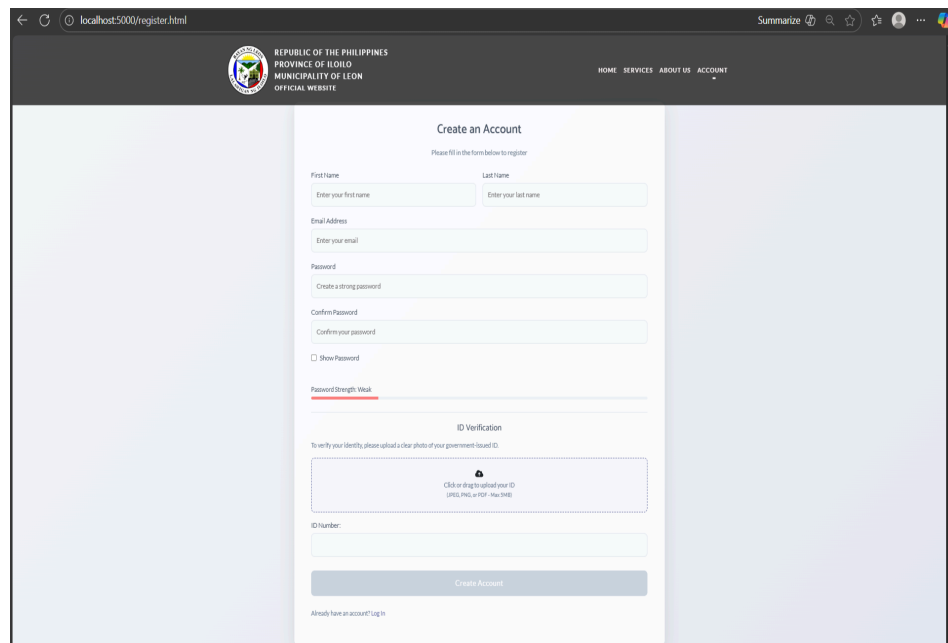
Figure 3.2 Landing Page

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Registration Form

Figure 3.3 The applicant or user will create an account.



The screenshot shows a web browser window displaying the 'Create an Account' registration form. The browser's address bar shows 'localhost:5000/register.html'. The website's header includes the logo of the Republic of the Philippines, Province of Iloilo, Municipality of Leon, and the text 'OFFICIAL WEBSITE'. Navigation links for 'HOME', 'SERVICES', 'ABOUT US', and 'ACCOUNT' are visible. The registration form itself is titled 'Create an Account' and includes a sub-header 'Please fill in the form below to register'. The form fields are: 'First Name' and 'Last Name' (both with 'Enter your first/last name' placeholder text), 'Email Address' (with 'Enter your email' placeholder text), 'Password' (with 'Create a strong password' placeholder text), and 'Confirm Password' (with 'Confirm your password' placeholder text). There is a 'Show Password' checkbox. Below these is a 'Password Strength: Weak' indicator. The 'ID Verification' section includes a note 'To verify your identity, please upload a clear photo of your government-issued ID.' and a button 'Click or drag to upload your ID (JPEG, PNG or GIF - Max 10MB)'. Below this is an 'ID Number' field. At the bottom of the form is a 'Create Account' button and a link 'Already have an account? Login'.

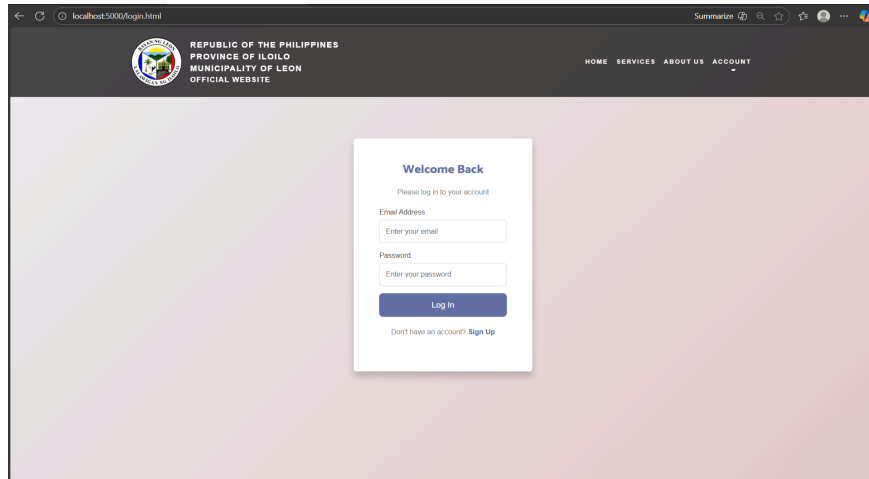
Figure 3.3 Account Registration

Login Form For Applicant/User

Figure 3.4 The page where a user will enter their credentials to gain access

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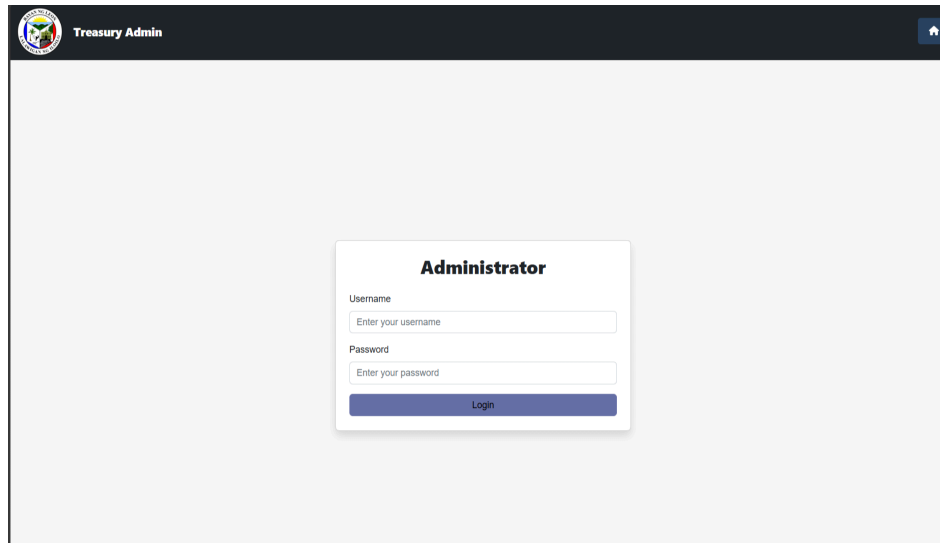


The screenshot shows a web browser window with the address bar displaying 'localhost:3000/login.html'. The page header includes the logo of the Municipality of Leon and the text 'REPUBLIC OF THE PHILIPPINES', 'PROVINCE OF ILOILO', 'MUNICIPALITY OF LEON', and 'OFFICIAL WEBSITE'. A navigation menu contains links for 'HOME', 'SERVICES', 'ABOUT US', and 'ACCOUNT'. The main content area features a login form with the title 'Welcome Back' and the instruction 'Please log in to your account'. The form has two input fields: 'Email Address' with the placeholder 'Enter your email' and 'Password' with the placeholder 'Enter your password'. Below these fields is a blue 'Log In' button. At the bottom of the form, there is a link that says 'Don't have an account? Sign Up'.

Figure 3.4 Login Form

Login Form for Admin

Figure 3.5 The page where an admin will enter their credentials to gain access to the Admin Dashboard.



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/treasury-admin/login.html'. The page header includes the logo of the Municipality of Leon and the text 'TREASURY ADMIN'. A navigation menu contains links for 'HOME', 'SERVICES', 'ABOUT US', and 'ACCOUNT'. The main content area features a login form with the title 'Administrator'. The form has two input fields: 'Username' with the placeholder 'Enter your username' and 'Password' with the placeholder 'Enter your password'. Below these fields is a blue 'Login' button.

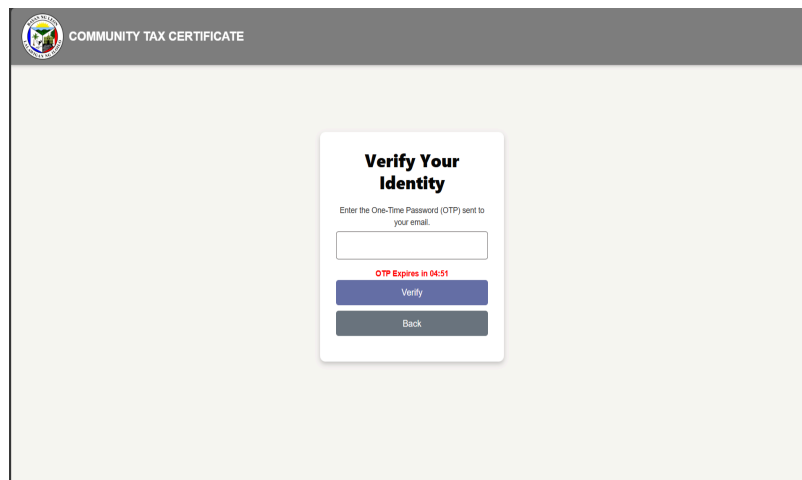
Figure 3.5 Login Form for Admin

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OTP Verification

Figure 3.6 A verification process for the user. It will send through email

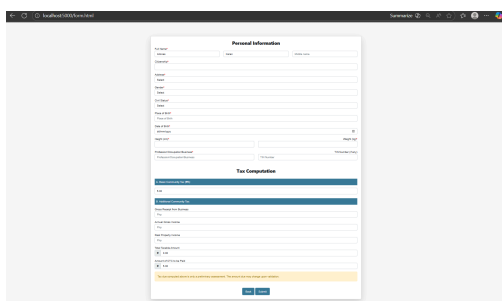


The screenshot shows a web interface for the 'COMMUNITY TAX CERTIFICATE' system. At the top left is a circular logo. The main content area is a light gray rectangle. In the center, there is a white card with the title 'Verify Your Identity' in bold. Below the title, it says 'Enter the One-Time Password (OTP) sent to your email.' There is a text input field for the OTP. Below the field, a red text label indicates 'OTP Expires In 04:51'. At the bottom of the card are two buttons: a blue 'Verify' button and a gray 'Back' button.

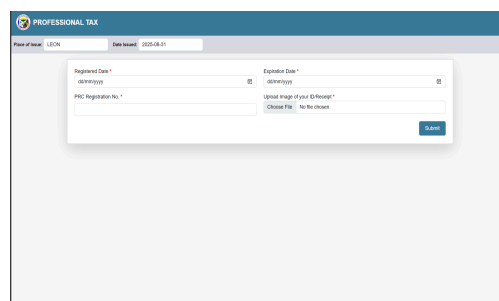
Figure 3.6 OTP Verification

Application Form for Cedula

Figure 3.7 This is where the user enter their credentials for applying a cedula



The screenshot shows a web browser window displaying a 'Personal Information' form. The form has multiple sections with labels like 'Last Name', 'First Name', 'Middle Name', 'Date of Birth', 'Sex', 'Civil Status', 'Religion', 'Address', 'Contact No.', and 'Email Address'. There are also sections for 'Tax Identification No.' and 'Professional Registration No.'. At the bottom, there is a 'Tax Computation' section with a table for 'Taxable Income' and 'Tax Amount'. A blue 'Next' button is at the bottom right.



The screenshot shows a web browser window displaying a 'PROFESSIONAL TAX' form. The form has a header with 'PROFESSIONAL TAX' and a 'Date Issued' field. Below the header, there are fields for 'Registered Date', 'Expiration Date', 'Professional Registration No.', and 'Upload Image of your ID/Photo'. There are 'Choose File' and 'No file chosen' buttons. A blue 'Next' button is at the bottom right.

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PROFESSIONAL TAX

Place of Issue: LEON Date Issued: 2025-08-31

Registered Date *
ddmm/yyyy

Expiration Date *
ddmm/yyyy

PRC Registration No. *

Upload Image of your ID/Receipt *
Choose File No file chosen

Submit

Figure 3.7 Application form for individual, professional and corporation

Admin Dashboard

Figure 3.8 This is where the admin can see the applicant records.

Cedula Admin Office

Leon's Admin Dashboard

Total Users: 8

Pending Applications: 3

Total Approved Cedula: 3

Total Rejected Cedula: 2

Pending Applications

No.	User ID	Full Name	Certificate Type	Date Request	Status	Action
53	81	Karen Allones	Individual Tax	2025-08-19	Pending	Approve Reject
11	454865	Karen Allones	Professional Tax	2025-08-15	Pending	Approve Reject
7	0	YG INCORPORATION	corporation Tax	0000-00-00	Pending	Approve Reject

Approved Cedula

No.	User ID	Full Name	Status	Action
47	80	Jeongwoo Park	Approved	Approve Reject
52	81	Karen Allones	Approved	Approve Reject

Rejected Cedula

No.	User ID	Full Name	Status	Action
49	59	Jiyong Karen	Rejected	Approve Reject

Figure 3.8 Admin Dashboard

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About Us

Figure 3.9 This where you can see the the details of a company

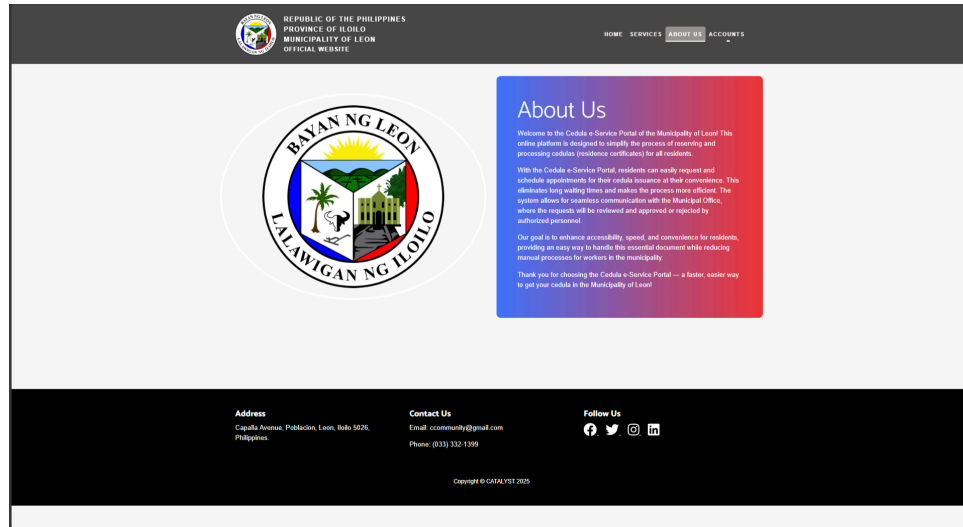


Figure 3.9 About Us

Use Case Diagram

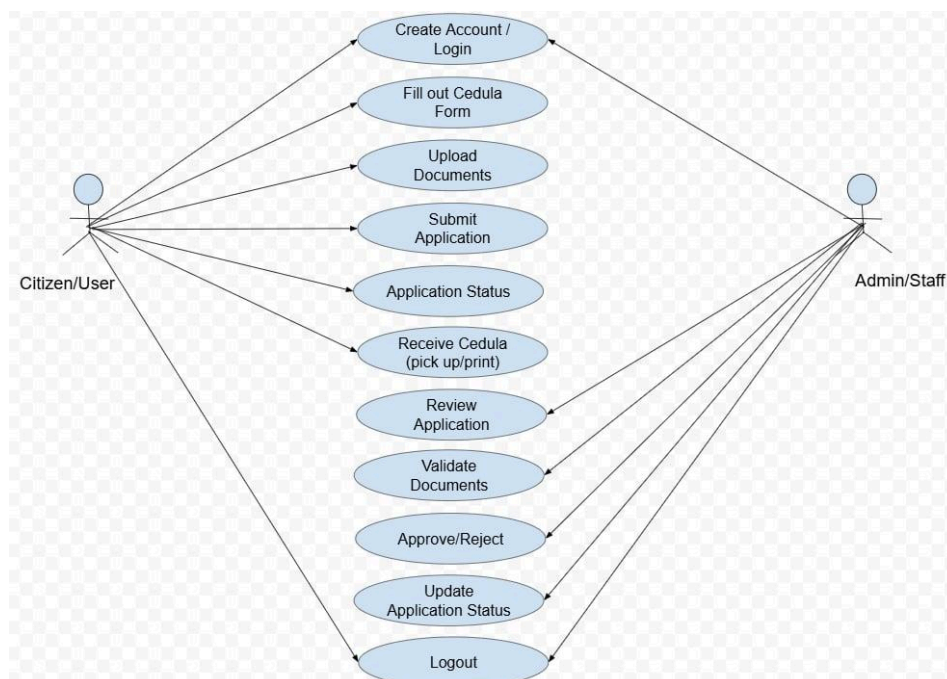


Figure 3.10 Use case diagram

The diagram shows that the client/user has a access on creating account, fill out application, upload documents, submit, receive a cedula application and log out, while the admin are the one that can access the admin Dashboard, review application, validate, approve/reject, update application status and log out.

Data Flow Diagram (DFD)

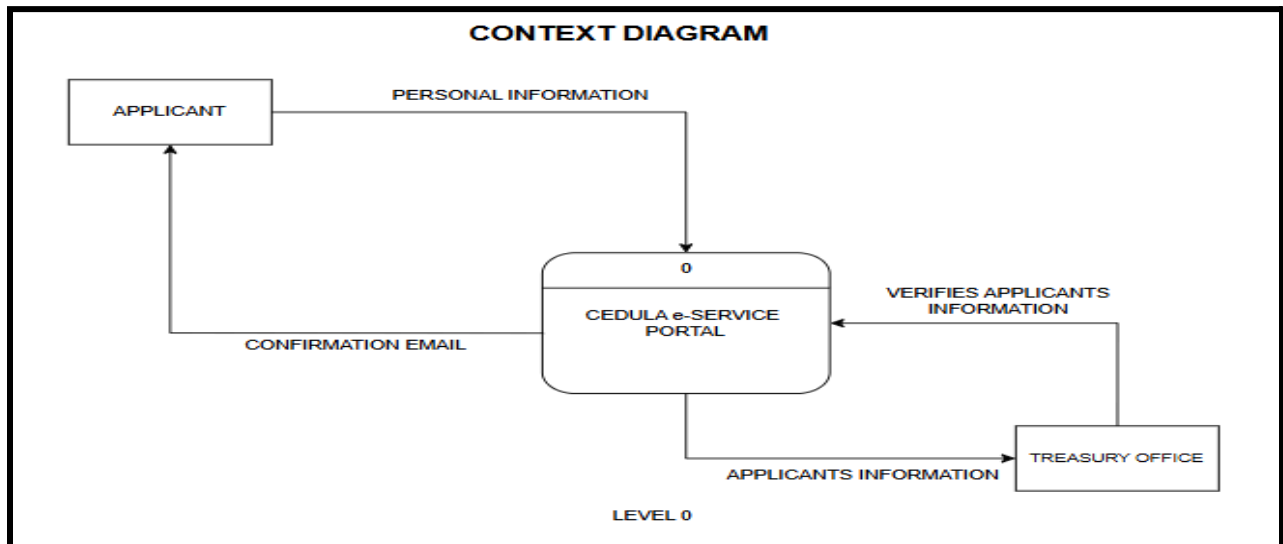


Figure 3.11 Level 0 Data Flow Diagram of Cedula e-Service Portal

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The context diagram illustrates the overall process flow of the Cedula e-Service Portal, showing how data moves between the system and its external entities. The process begins when the applicant submits their personal information to the Cedula e-Service Portal. The system then transmits the applicant's details to the Treasury Office for verification. Once the Treasury Office verifies the information, it sends the results back to the portal. After successful verification, the portal sends a confirmation email to the applicant, signaling the completion of the transaction. This diagram provides a high-level view of the interaction between the applicant, the Cedula e-Service Portal, and the Treasury Office, emphasizing the exchange of information required to process a cedula application.

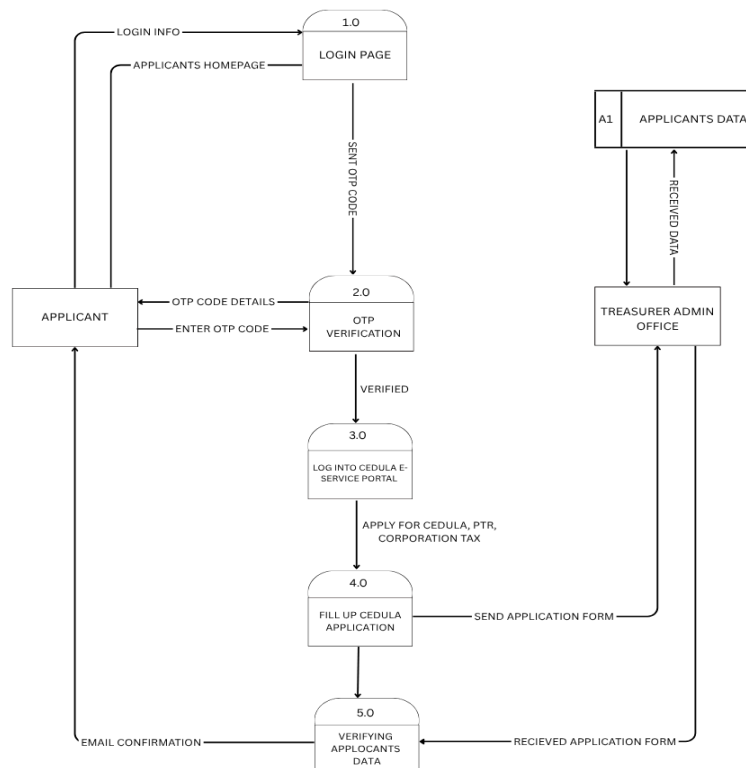


Figure 3.12 Level 1 Data Flow Diagram of Cedula e-Service Portal

The Level 1 Data Flow Diagram illustrates the detailed process of the Cedula e-Service Portal from user login to final verification. The process begins when the applicant enters their login credentials on the login page, which then generates and sends a One-Time Password (OTP) code. The applicant inputs the OTP into the OTP verification process, and once verified, they gain access to the Cedula e-Service Portal. From there, the applicant logs into the portal and selects the option to apply for a cedula by filling out the online application form with the required personal details. The completed form is sent to the Treasury Admin Office, which receives the applicant's data and forwards it for verification. The verification process checks the accuracy and validity of the applicant's information. Once verified, the system sends a confirmation email to the applicant, signaling the successful completion of the cedula application process. This diagram emphasizes the step-by-step interaction between the applicant, the system, and the treasury office, highlighting the secure login, verification, and data handling procedures.

Entity Relationship Diagram

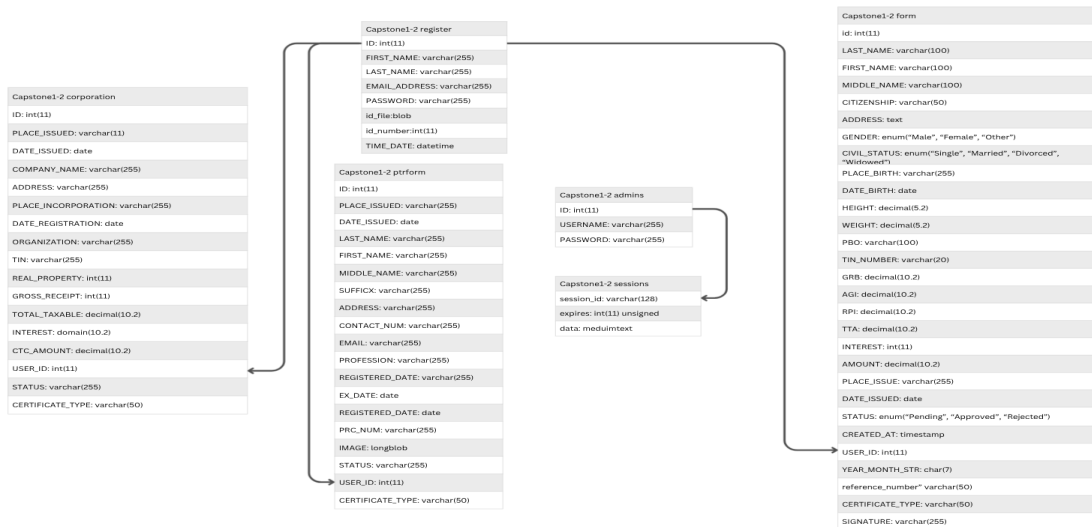


Figure 3.13 Entity Relationship Diagram of Cedula e-Service Portal

Security Consideration

The security mechanism implemented in the cedula e-service portal are the following:

- 1. User Authentication:** Implement multifactor (MFA) authentication using otp (one time password) via email to verify user identity.
- 2. Role-Based Access Control:** Define user roles with specific permission to access system resources and functionalities.

3. Data Protection: Encrypt sensitive data that the user input. This includes personal information, financial data and authentication credentials.

Technology Stack

These are the list of all the tools, platforms, and technologies used in the development:

- Frontend: Bootstrap, HTML, CSS
- Backend: NodeJs, Express Js
- Database: MySQL

Data Gathering Techniques

Survey Questionnaire for System Assessment

The survey is designed to gather feedback from users and the admin regarding their satisfaction from using the "Cedula e-Service Portal". The data collected from this survey helped in designing a system that meets the needs of users and addresses existing challenges in observation. Please rate the following statements based on your experience with the system: (Very Satisfied, Satisfied, Neutral, Dissatisfied, Very Dissatisfied and a open ended feedback)

1. Overall Satisfaction: How satisfied are you with the overall experience of using the Cedula e-Service Portal?

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- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

2. Ease of Use: How easy was it to use the Cedula e-Service Portal?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

3. Efficiency: How satisfied are you with the efficiency of the Cedula e-Service Portal in processing your application?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

4. Security: How confident are you in the security of the Cedula e-Service Portal?

- ☐ Very Satisfied

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- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

5. Online Application Submission: How satisfied are you with the online application submission process?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

6. Real-time Application Tracking: How satisfied are you with the real-time application tracking feature?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

7. OTP Verification: How satisfied are you with the OTP (One-Time Password) verification process?

- ☐ Very Satisfied
- ☐ Satisfied

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- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

8. Appointment Scheduling: How satisfied are you with the appointment scheduling system for on-site processing (payment and thumbprint verification)?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

9. Automated Notifications: How satisfied are you with the automated notifications you received regarding your application status and appointments?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

10. Confirmation Email: How satisfied are you with the confirmation email and verification link process for scheduled appointments?

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- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

11. Digital Literacy (If Applicable): If you required training or support to use the portal, how satisfied were you with the assistance provided?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

12. Accessibility: How satisfied are you with the accessibility of the Cedula e-Service Portal, especially if you have limited mobility or reside in a remote area?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

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13. Comparison to Traditional System: Compared to the traditional paper-based Cedula process, how satisfied are you with the e-Service Portal?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

Open-Ended Feedback:

1. What did you like most about using the Cedula e-Service Portal?
2. What improvements would you suggest for the Cedula e-Service Portal?
3. Do you have any other comments or suggestions?

Chapter IV

Results and Discussion

This chapter presents the outcome of the development in the testing phases of the cedula e-service portal. It interprets the key findings and evaluates whether the system meets its intended objective, such as streamlining the process of cedula. It analyzes its effectiveness based on the research goals and user feedback. This will also reflect how the system addresses the problems identified in earlier chapters and supports the project overall goals

Summary of the Data Sources and Analysis Methods Used

The data used for analysis in this was collected from different sources including system logs, records and user testing section. The main data source was the portal database which stores user credential, application records and the recorded feedback during testing. Other information was gathered through survey and with collaboration of the client/user to evaluate the system performance, usability and for the improvement of the system.

Both quantitative and qualitative methods were used in analyzing the data. Quantitative data such as application processing times and errors rates, was examined to measure efficiency improvements. Meanwhile, qualitative feedback from users was organized and interpreted to evaluate the portal usability, design and reliability providing a comprehensive understanding of its effectiveness.

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Task Manager

Gantt Chart / Project Timetable - Cedula e-Service Portal

Week	Task	Programmer	UI/UX Designer	Documentor
1	Project planning & requirement gathering	✓	✓	✓
2	Research & system design planning	✓	✓	✓
3	Use Case & DFD creation		✓	✓
4	Database design & structure creation	✓		✓
5	UI wireframes & prototype mockups		✓	
6	Frontend development begins	✓	✓	
7	Backend	✓		

	development begins			
8	Integration of frontend and backend	✓	✓	
9	System testing & debugging	✓	✓	
10	User testing & feedback collection	✓	✓	✓
11	Final revisions (code, UI, documentation)	✓	✓	✓
12	Documentation finalization & formatting		✓	✓
	Final project submission	✓	✓	✓

Legend:

-  = Active role/task in that week

Presentation of Results

The cedula e-service portal was designed, developed and tested to automate and simplify the process of obtaining cedula. The testing phase will focus on evaluating the system performance log. The finding revealed that the system improved the speed and accuracy of cedula processing compared to the manual paper-based method. These outcomes confirm that the cedula e-service portal effectively meets its goal of enhancing processing and supporting efficient service delivery.

Chapter V

Summary, Conclusion, and Recommendation

Summary

This study addressed the inefficiencies of the traditional Community Tax Certificate (CTC) or Cedula application process, which often involves time-consuming procedures. The project aimed to design, develop, and implement a user-friendly and secure Cedula e-Service Portal to modernize and streamline the cedula application and issuance process. The e-Service Portal leverages online application submission, OTP verification, and appointment scheduling to improve efficiency, accessibility, and security for citizens and government agencies. Agile methodology was chosen to be the SDLC, as the client has requested making changes throughout the development of the system. A three-tier architecture system that covers from the front end, to backend database of the project. The portal database stores user credentials, application records, and recorded feedback during testing and user interviews were conducted. User interviews were conducted to further the efficiency of the project. Through the use of Quantitative data such as logs and Qualitative data user feedback was also taken for the further improvement and assessment of the project.

Conclusion

The Cedula e-Service Portal offers a promising solution to modernize the outdated Cedula application process, the e-service portal will give way for the future of the municipality to further enhance the digitalization and modernization for the municipality of Leon. By enabling online application submission, integrating secure OTP verification, and streamlining appointment scheduling, the portal addresses key issues such as time-consuming processing, limited accessibility, and security concerns associated with the traditional paper-based system. Implementation and testing have demonstrated the system's potential to significantly reduce processing times, enhance accessibility for citizens in remote areas or with mobility limitations, and improve the overall security of the Cedula application process. User feedback has helped enhance the project as well. Based on the results, the successful development and initial implementation of the Cedula e-Service Portal highlights the potential for technology to transform government services, improve citizen engagement, and enhance administrative efficiency.

Recommendations

To maximize the benefits of the Cedula e-Service Portal and ensure its long-term success, the following recommendations are proposed:

- **Address Digital Literacy:** Implement training programs and provide user support resources to assist citizens with limited digital skills in using the online portal effectively. The integration of tutorials in the system might further improve the system.
- **Improve Internet Access:** Explore partnerships with local internet providers or community centers to provide reliable internet access to citizens, particularly in remote areas.
- **Enhance System Integration:** Continue to integrate the e-service portal with other relevant government systems and databases to streamline data sharing and improve operational efficiency.
- **Optimize On-Site Processing:** Evaluate the efficiency of the on-site processing component (payment and thumbprint verification) and explore options for further streamlining this aspect of the process, potentially through the use of mobile payment solutions or biometric authentication technologies.

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- **Prioritize Security:** Continuously monitor and update security measures to protect against evolving cyber threats and ensure the privacy and security of citizen data.
- **Gather Ongoing Feedback:** Establish a mechanism for collecting ongoing user feedback to identify areas for improvement and ensure that the e-service portal continues to meet the evolving needs of citizens and government agencies.
- **Expand Functionality:** Explore opportunities to expand the functionality of the e-service portal to include other government services, creating a one-stop-shop for citizens' needs.